

Fraud Detection System: Architecture & Model Logic

1. Project Overview

This document outlines the workflow and model logic for a Real-Time Fraud Detection System. The system transitions from rigid, rule-based heuristics to an adaptive anomaly-detection framework driven by unsupervised machine learning. By profiling normal transactional behavior based on data density and spatial proximity, the architecture isolates high-risk deviations.

2. Feature Engineering Pipeline

The dataset undergoes transformations to extract high-variance statistical and behavioral metrics:

- **Temporal Dynamics:** Calculating time-deltas such as 'DaysSinceLastTransaction' to capture frequency anomalies.
- **Historical Ratios:** Generating 'AmountVSHistoryRatio' to evaluate current transaction magnitudes against historical averages.
- **Directional Multipliers:** Encoding transaction types (Credit/Debit) into a 'SignedAmount' space to map vector directionality.

3. Unsupervised Machine Learning Logic

The core detection engine employs a multi-model framework to evaluate transactions from different algorithmic perspectives.

Isolation Forest (Tree-Based)

Isolation Forest identifies anomalies by randomly selecting a feature and a split value. Since anomalies are 'few and different', they require fewer splits to be isolated. The model assigns an anomaly score based on the path length; shorter paths indicate higher fraud probability.

Local Outlier Factor - LOF (Density-Based)

LOF computes the local density deviation of a given transaction relative to its neighbors. Transactions that have a substantially lower density than their neighbors are flagged as outliers. This relies on distance metrics to identify structural deviations in the transaction space.

Reasoning for using both the models:

I used both because financial fraud happens on two different scales. Isolation Forest acts like a macro-filter that instantly catches loud, massive global anomalies like huge spending spikes. Local Outlier Factor acts like a micro-microscope that looks at local neighborhoods to catch quiet deviations, like account takeovers on dormant accounts. Combining them allows us to build a tiered defense—automatically blocking high-confidence threats where both models agree, while routing single-model warnings to a dashboard to prevent false-alarm fatigue.

4. Interactive Dashboard & GenAI Integration

To operationalize the model outputs, the system features a Gradio-based interactive dashboard. Analysts can select specific flagged transactions via a dropdown interface and retrieve detailed transaction data. A conceptual Retrieval-Augmented Generation (RAG) module parses the transaction context and ML decision boundaries to output natural language explanations, bridging the gap between raw statistical anomalies and human-readable risk assessments.